Phase 2 - Requirement Analysis

Project: ComicCraft - AI Comic Story Creator **Team ID / members:** Fill in your team's details **Date:** Fill in submission date

User journey

Stage	User action and touchpoint	Need / feeling	System response
Start	Opens ComicCraft home page	Wants an obvious starting point	Shows prompt form and available options
Describe	Enters story, hero, setting, tone, art style	Wants the story to reflect their idea	Validates submitted fields
Generate	Submits the form	Expects progress and a result	Builds a five-panel story and illustrations
Review	Reads panel titles, scenes, and dialogue	Wants to check story flow	Displays comic preview
Export	Selects PDF download	Wants a portable copy	Generates and returns a PDF

Functional requirements

ID	Requirement	Priority	Implementation status
FR-01	Accept story prompt, hero, setting, tone, art style, and image option	High	Implemented in web form and request schema
FR-02	Generate a structured five-panel story using Gemini when configured	High	Implemented; demo fallback is available without Gemini
FR-03	Generate panel illustrations through Hugging Face when configured	High	Implemented as optional service; local SVG fallback is available
FR-04	Show panel-by-panel descriptions and dialogue in the browser	High	Implemented in preview template
FR-05	Export the comic as a PDF	High	Implemented
FR-06	Offer JSON comic-generation endpoint	Medium	Implemented at `POST /generate-comic/json`
FR-07	Report service configuration and application health	Medium	Implemented at `GET /health`
FR-08	Keep API credentials outside source code	High	`.env` configuration supported; never commit `.env`

Non-functional requirements

- **Usability:** Clear form labels, readable panel preview, responsive pages.
- **Reliability:** If a cloud model is unavailable, use the documented local fallback where supported and log server-side errors.
- **Security:** Store keys in a local `.env`; do not show or commit secret values.
- **Maintainability:** Keep routes, schemas, and generation services in separate Python modules.
- **Performance:** This prototype has no load-test result. Measure request time and concurrent use before claiming production capacity.
- **Accessibility:** Use descriptive labels and meaningful image alt text; verify keyboard navigation before public deployment.